

40-Day Summer Internship Diary

Student: Harsh Tiwari

Branch: Information Technology

Technology: Generative AI

Day 1

Date: 08 June 2026

Topic: Introduction to Generative AI

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 2

Date: 09 June 2026

Topic: Python & Environment Setup

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 3

Date: 10 June 2026

Topic: Fundamentals of NLP

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 4

Date: 11 June 2026

Topic: Text Preprocessing

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 5

Date: 12 June 2026

Topic: TF-IDF & Feature Extraction

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 6

Date: 15 June 2026

Topic: Word Embeddings

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 7

Date: 16 June 2026

Topic: Cosine Similarity

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 8

Date: 17 June 2026

Topic: Artificial Neural Networks

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 9

Date: 18 June 2026

Topic: Convolutional Neural Networks

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 10

Date: 19 June 2026

Topic: Recurrent Neural Networks

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 11

Date: 22 June 2026

Topic: Long Short-Term Memory

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 12

Date: 23 June 2026

Topic: GRU Architecture

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 13

Date: 24 June 2026

Topic: Attention Mechanism

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 14

Date: 25 June 2026

Topic: Transformers

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 15

Date: 26 June 2026

Topic: Encoder-Decoder

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 16

Date: 29 June 2026

Topic: BERT & GPT

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 17

Date: 30 June 2026

Topic: Large Language Models

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 18

Date: 01 July 2026

Topic: Prompt Engineering

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 19

Date: 02 July 2026

Topic: Vector Databases

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 20

Date: 03 July 2026

Topic: RAG

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 21

Date: 06 July 2026

Topic: Fine-Tuning

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 22

Date: 07 July 2026

Topic: Dataset Preparation

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 23

Date: 08 July 2026

Topic: Project Planning

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 24

Date: 09 July 2026

Topic: System Architecture

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 25

Date: 10 July 2026

Topic: Data Preprocessing

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 26

Date: 13 July 2026

Topic: Model Training

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 27

Date: 14 July 2026

Topic: Model Evaluation

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 28

Date: 15 July 2026

Topic: FastAPI

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 29

Date: 16 July 2026

Topic: Streamlit

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 30

Date: 17 July 2026

Topic: SQLite

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 31

Date: 20 July 2026

Topic: Online Verification

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 32

Date: 21 July 2026

Topic: Entity Matching

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 33

Date: 22 July 2026

Topic: Deployment

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 34

Date: 23 July 2026

Topic: Testing

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 35

Date: 24 July 2026

Topic: Debugging

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 36

Date: 27 July 2026

Topic: Project Integration

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 37

Date: 28 July 2026

Topic: Documentation

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 38

Date: 29 July 2026

Topic: Presentation Preparation

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 39

Date: 30 July 2026

Topic: Final Review

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.

Day 40

Date: 31 July 2026

Topic: Project Demonstration

Studied the topic, implemented practical examples in Python/Jupyter Notebook, performed hands-on coding, analyzed outputs, resolved implementation errors, and documented the learning outcomes.

Learning: Improved understanding of Generative AI concepts and practical implementation.